

content of the speaker's meaning. In our example it is the set of assumptions listed in (3)². In other words, Mary's informative intention is to increase the likelihood that Peter will mentally represent these assumptions as Mary's intended meaning, ideally even so much that he believes them.

The effect of Mary's communicative intention is to make it (more) manifest to Peter not only that she intends to make the assumptions in (3) (more) manifest to him, but also the fact that Mary intends Peter to recognise that she intends to do so. In other words, a communicator making her informative intention mutually manifest ensures that this intention of hers becomes overt to the audience. Overtly intentional acts are processed in a special way by organisms with minds that strive for efficiency in information processing and knowledge representation, as we shall see shortly in the following sections.

3.2 Inputs to cognitive processes compete for relevance

Sperber and Wilson (1995, pp. 261–262) point out that as a biological function, cognition is under pressure to process inputs *efficiently*. To measure the efficiency of input processing, relevance theory introduces the notion of *relevance* as a technical term. An input is relevant to the extent that it leads to the most *positive cognitive effects* for the least processing effort involved in deriving these effects. Positive cognitive effects are improvements of the mind's representation of the world, and may include confirmation (strengthening) of existing knowledge, deletion of assumptions that contradict existing knowledge but are false (contradiction and elimination), or the acquisition of new information as a result of combining the input with existing assumptions (contextual implication; Sperber and Wilson 1995, pp. 118–132).

Suppose you are sitting at your desk writing a paper. You pause and look at the wall in order to give your eyes rest from looking at the computer screen. There happen to be too small black dots at the wall. You hardly notice them and most likely don't pay attention to them. Then suddenly you have the impression that one of the dots moves around ever so slightly. You sit up and look more closely at this dot. Why? Because the input THE DOT MAY HAVE MOVED easily gives rise to the following thought chain: if it indeed has moved, it may be a living thing, an insect, for instance a spider. And if it is a spider, you don't want it to be there, you want to remove it. In other words: the input has become more relevant to you: there are easily available cognitive effects.

At the same time, you stop paying attention to the other dot that hasn't moved. But you do get up, grasp a sheet of paper and go to the wall to check. Which dot will you investigate first? Surely the dot that you think has moved. Why not the other? After all, this other dot is of roughly the same size and shape. But it hasn't given rise to more cognitive effects as the other dot has.

This example illustrates a descriptive generalisation that relevance theory makes about cognition in general: the mind tends to pay attention to inputs that are most relevant. Sperber and Wilson (1995, p. 260) put it in these terms:

- (7) Human cognition tends to be geared to the maximisation of relevance.

By allocating scarce processing resources primarily to such inputs that promise to be most relevant, information is processed efficiently. Sperber and Wilson (2015, pp. 134–135) point out that there is a close connection between the notions of manifestness and of relevance:

²In fact, this is a simplified statement that will be revised shortly

However, manifestness is this *ceteris paribus* probability of influence, rather than the factors that contribute to it. The same point can be made on the basis of the Cognitive Principle of Relevance ('Human cognition tends to be geared to the maximisation of relevance'): Given a belief which has a cognitive effect and which is therefore relevant in a categorical sense, its relevance will be comparatively greater 1) to the extent that its processing is less costly because it is more salient, and 2) to the extent that its epistemic strength is greater. If a proposition is relevant at all, then the greater its manifestness, the greater its relevance.

3.3 Overtly intentional inputs claim a good measure of relevance

Overtly intentional inputs are relevant in terms of the intentions that the communicator expects the audience to infer. This means essentially that once the audience has grounds to attribute a communicative intention to a communicator, it is forced to put in the effort to identify an informative intention. Given that the mind tends to allocate processing resources for those inputs that are most relevant, this means that overtly intentional inputs raise the expectation that they are relevant to a specific degree: they must at least be relevant enough for the audience to be worth the attention. But given the mind's tendency to maximise relevance, it will be beneficial for the communicator if the overtly intentional input is more relevant to the audience than this minimum requirement. On the other hand, rational audiences can not expect perfection: communicators may be limited in their *abilities* and *preferences*: in regard to their abilities, communicator's may not think of the best words, the easiest grammatical expression, or simply have an erroneous assumption of what is relevant for the audience. In regard to their preferences, communicators may want to conform to societal expectations, respect cultural taboos, and so on. All in all, overtly intentional inputs promise to be *optimally relevant*, where *optimal relevance* is a level of relevance with a lower bound of being relevant just enough to be worth the attention, and an upper bound of being as relevant as the communicator's abilities and preferences allow. This is expressed as the communicative principle of relevance:

- (8) Communicative Principle of Relevance
Every [act of overtly intentional communication] conveys a presumption of its own optimal relevance. (Slightly adapted from Wilson and Sperber 2004, p. 612)

3.4 Processing overtly intentional inputs with an efficient heuristic procedure

Because overtly intentional inputs come with an implicit claim that they are optimally relevant to the audience, the informative intention underlying them can be found following the heuristic procedure in (9):

- (9) *The relevance-theory processing heuristic*
 - a. Access the most easily accessible interpretive hypothesis, consisting of a hypothesis about the implicit import, a hypothesis about the fine-tuning

of explicit meaning, together with the contextual assumptions that support these hypotheses.

- b. Check if the stimulus, under this interpretation, yields enough cognitive effects of the right kind to meet the audience's expectation of relevance.
- c. If it does meet relevance expectations, the audience is justified to assume that this is the interpretation intended by the communicator.
- d. If it does not meet relevance expectations, the audience may access other interpretive hypotheses in the order of accessibility until an interpretation satisfying relevance expectations is found, or the search is abandoned because the processing effort is deemed too high.

Let us apply this processing heuristic to example (1b) and see how this procedure explains how the audience arrives spontaneously and intuitively at the interpretations in (3).

3.5 Processing implicatures

At the time Mary in example (1b) is taking her turn in the conversation, Peter (the audience) has specific expectations about the content of cognitive effects that any utterance of Mary must have in order to be relevant to him: it must be interpretable as providing a positive or negative answer to the question Peter asked, whether Mary wants to have some coffee. However, the utterance itself does not say anything about coffee itself. But it does convey the concept of **ENERGIZING DRINKS**. This concept bundles our encyclopaedic knowledge about energizing drinks, which presumably includes a list of examples:

- (10) **ENERGIZING DRINKS**
...
Coffee is an energizing drink.
Cola is an energizing drink.
Redbull is an energizing drink.
Tea is an energizing drink.
Peppermint tea is an energizing drink.
...

Since the concept **COFFEE** has been activated by the previous conversational turn (Peter's question), the assumption **COFFEE IS AN ENERGIZING DRINK** is more highly activated, and thus easier accessible than, the other pieces of encyclopaedic information associated with the concept **ENERGIZING DRINKS**. Assuming that this idea is intended to be used as context, the utterance can be interpreted along the lines in (11):

- (11) a. Coffee is an energizing drink.
b. If coffee is an energizing drink, Mary does not want to drink coffee.
c. Mary does not want to drink coffee.

Accessing assumption (11a) and using it as context, this assumption can be used as second premise in a syllogism using the conditional (11b) to support the conclusion in (11c). This conclusion in turn provides a cognitive effect of the expected kind. Since these interpretive hypotheses are very easy to access, and the utterance achieves